

The question

Can we rebuild the contacts-v1 training corpora with a decontamination pass that actually covers the eval set we report on — and what does it cost?

Issue #213 measured the overlap: 323 of the 554 eval proteins (58%) have a significant training homolog. This is the other half — the fix, and its price.

What is filtered today

Two corpora, two stories.

AFDB (4,129,682 documents, #53) has NO eval decontamination at all. Its split is afdB-24M's own cluster hash, so nothing leaks across the AFDB split, but the 554 eval proteins were never purged from it. #41 found 99 of 100 FoldBench monomers fall in a fold the model trained on; those verdicts became reporting strata and nothing else.

ESM-Atlas (66,759,922 documents, #91 then #139) has one filter, with four gaps. It dropped 41,517 sequences at 40% identity or above, but: FoldBench-100 was never in the reference, which is 18% of the benchmark; 40% is looser than our own 30% bar; the filter is sequence-only, on the axis #41 showed is the wrong one; and #91's own stated follow-up never happened.

The design

Both corpora are already materialised, so decontamination is a row filter on entry_id, not a regeneration. Three tiers, priced before anything is retrained:

A — sequence. Identity at or above 30% over at least half the eval protein, or E at or below $1e-3$. B — plus structurally redundant. A, plus TM 0.90 or above to any eval structure. C — plus fold-level purge. B, plus whole clusters at TM 0.50 or above.

The expensive inputs already existed: #213's 70.9M-sequence MMseqs2 database, and #41's Foldseek database of the 1.33M AFDB cluster representatives. No new pyconfind, no document rebuild, no cluster job.

What we found

Tier A is cheap: 1.89% of AFDB, 1.57% of ESM-Atlas. And it is not an artefact of a threshold — sweeping the search's reporting depth over six decades moves it by a quarter of a percentage point.

All four of #91's gaps are real. Of the surviving contaminated ESM-Atlas rows, 47% sit in the 30-40% band, 24% are remote homologs below any identity bar, 17% are reachable only from FoldBench-100 — and 12% (123,713 rows) clear #91's own 40%/50% rule. That last group is the measured price of its -s 4.0 search sensitivity, a trade-off its own script flagged in a comment.

#41 was right about the axis, and acting on it is cheap at Tier B: 22,320 AFDB documents are structurally near-identical to an eval structure while being invisible to any sequence search. That is 0.54% of the corpus.

Tier C is unaffordable: 37.31% of AFDB, and 95% of that is structure-only. The reason is not an unlucky threshold. The mode of "best TM to any of the 554" sits essentially at the 0.5 same-fold boundary — a third of AFDB's structural clusters simply are the same fold as something in a 554-protein eval set.

Three quarters of that cost protects the 396 de novo designs. Scoping the fold purge to the 158 natural eval proteins costs 9.38% instead of 36.77%. Designs are small idealised bundles that share a fold with an enormous share of AFDB, and they are the proteins with no evolutionary relatives to leak through in the first place.

Where it lands

H1 holds through Tier B; H0 holds for Tier C. The recommendation is to publish Tier B for AFDB and Tier A for ESM-Atlas, and to decline Tier C with 37.31% as the number that justifies declining it. The 9.38% natural-only variant is a real middle option if fold novelty later becomes load-bearing.

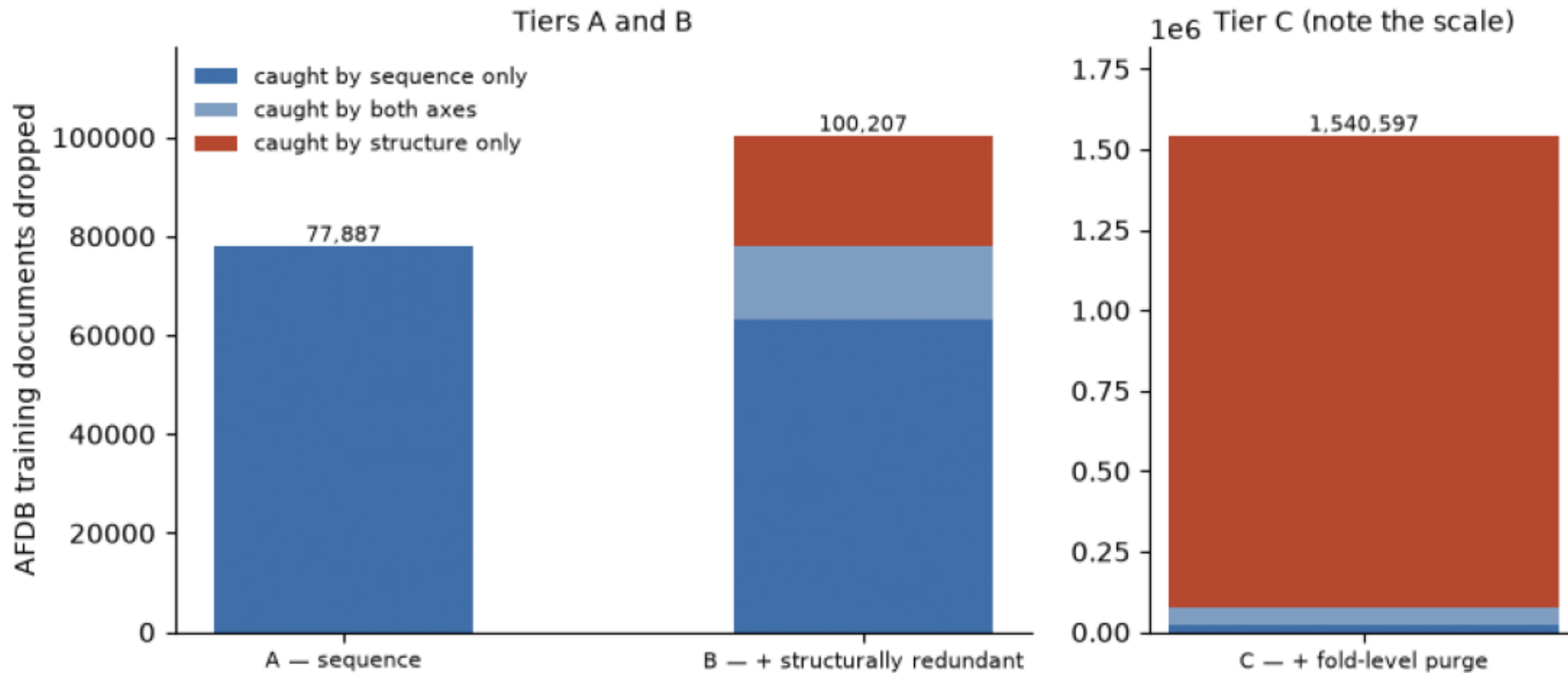
Do not pay for the ESM-Atlas Foldseek build (about \$1k) yet. It was gated on this table. The only tier it could serve is B, since C is declined, and on the arm we can measure, B's structural increment is 0.54%. Revisit if the retrain moves. The caveat: ESM-Atlas is metagenomic, so its fold distribution against this eval set need not match AFDB's.

Stage 5 — retraining the #199 recipe on the decontaminated mixture and re-scoring against #213's homology-free subset — is the actual test of H1. Nothing here measures accuracy.

axis_decomposition.png

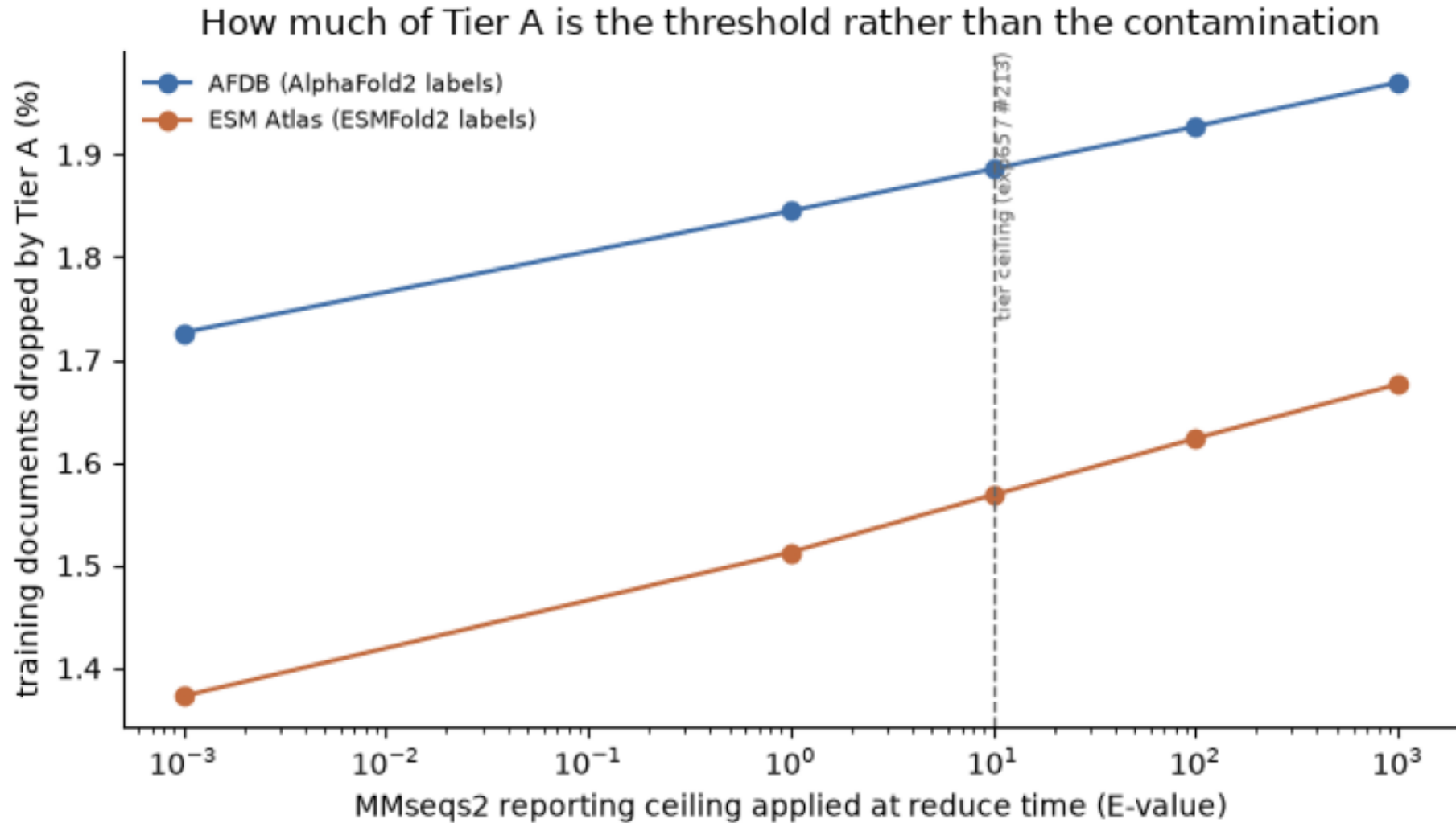
Red = what #41 predicted and nobody had measured — documents a sequence filter keeps and a structural one removes: 22,320 at Tier B, 1,462,710 at C.

Which axis is doing the work (AFDB arm)



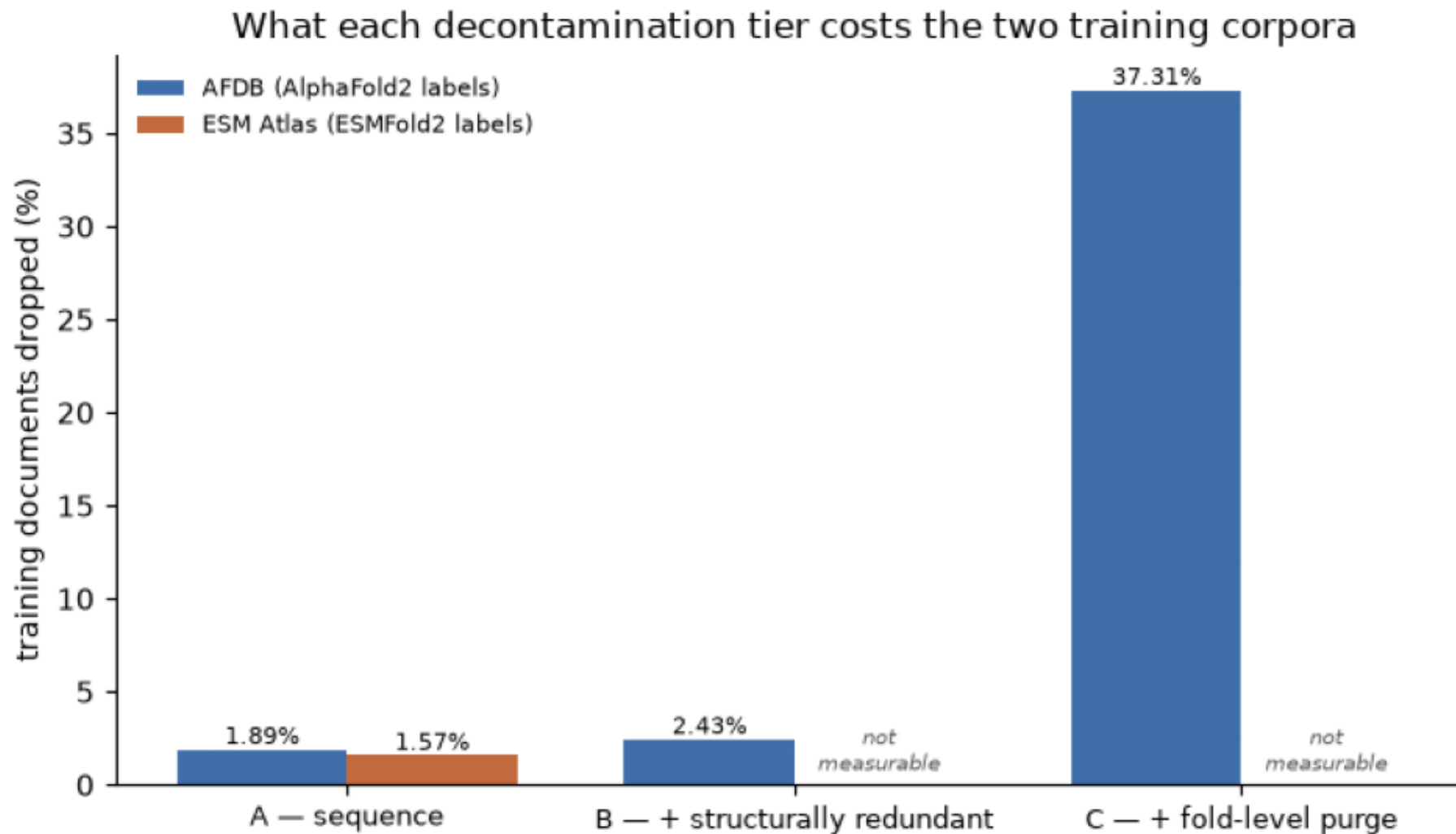
evaluate_sensitivity.png

The E-value arm of Tier A contributes nothing above $1e-3$; everything the curve gains after that is the identity arm, which has no significance floor of its own.



survival_by_tier.png

Share of each published train corpus removed at each tier. ESM-Atlas has no structural database, so its Tier B/C cells are unmeasured rather than zero — building one is the job this table gates.



tm_distribution.png

945,861 of the 1,304,911 AFDB train representatives got a hit and are counted here; the other ~359k had none and are off the left edge.

